

Challenge (Habanero)

- Q1.** A recipe for making cookies requires $2x$ cups of flour and $3x$ cups of sugar. If a baker has 12 cups of flour and 15 cups of sugar, how many batches of cookies can be made?
(A) 2 batches
(B) 3 batches
(C) 4 batches
(D) 5 batches
(E) 6 batches
- Q2.** A construction site requires x hours to complete the foundation and $2x$ hours to build the frame. If the total time available is 18 hours, what is the maximum value of x ?
(A) 4 hours
(B) 5 hours
(C) 6 hours
(D) 7 hours
(E) 8 hours
- Q3.** A chef is making a sauce that requires x cups of tomato puree and $x + 2$ cups of water. If the total amount of liquid is 12 cups, what is the value of x ?
(A) 3 cups
(B) 4 cups
(C) 5 cups
(D) 6 cups
(E) 7 cups
- Q4.** A builder is constructing a house with a rectangular base. The length of the base is x meters, and the width is $x - 2$ meters. If the area of the base is 24 square meters, what is the value of x ?
(A) 4 meters
(B) 5 meters
(C) 6 meters
(D) 7 meters
(E) 8 meters
- Q5.** A recipe for making bread requires x cups of flour and $x - 1$ cups of yeast. If a baker has 15 cups of flour and 10 cups of yeast, how many batches of bread can be made?
(A) 5 batches
(B) 6 batches
(C) 7 batches
(D) 8 batches
(E) 9 batches
- Q6.** A construction crew can build x meters of fence in one day. If they work for 5 days, they can build $x + 10$ meters of fence. How many meters of fence can they build in one day?
(A) 5 meters
(B) 6 meters
(C) 7 meters
(D) 8 meters
(E) 9 meters
- Q7.** A chef is making a dish that requires x cups of oil and $2x$ cups of vinegar. If the total amount of liquid is 18 cups, what is the value of x ?
(A) 4 cups
(B) 5 cups
(C) 6 cups
(D) 7 cups
(E) 8 cups
- Q8.** A builder is constructing a bridge with a length of x meters and a width of $x - 3$ meters. If the area of the bridge is 30 square meters, what is the value of x ?
(A) 6 meters
(B) 7 meters
(C) 8 meters
(D) 9 meters
(E) 10 meters

Open Ended Questions (FRQ)

- Q9.** A bakery is making a special cake that requires x cups of flour and y cups of sugar. The baker has 20 cups of flour and 15 cups of sugar. Write a system of equations to represent the situation and solve for x and y if the cake requires $2x + 3y = 45$ and $x + 2y = 20$. Show your work and explain your answer.
- Q10.** A construction company is building a new skyscraper with x floors and y elevators. The company has a budget of 10 million and each floor costs 0.5 million to build, while each elevator costs 1 million. Write a system of equations to represent the situation and solve for x and y if the company has $2x + 5y = 50$ and $x - 2y = -10$. Show your work and explain your answer.

Chef's Tips

- Read each question carefully before answering
- Show all work and explain answers for free response questions
- Use a calculator only when necessary